

A *sequence* is a data structure that stores a collection of items in a specific order. In some languages, a sequence can only contain items of one type; in other languages, including Python, a sequence can have items of different types.

Sequences are important when you need to store items in a specific order. For example, a list of website users might be ordered according to when each user registered.

With a sequence, you can do the following:

- Add and remove items
- Work with individual items or groups of items
- Determine whether a value is in the sequence
- Look for duplicate or unique items
- Loop through the sequence and do something with each item
- Work with items that match certain conditions.

In Python, the main sequence type is the *list*. Lists are *mutable*, meaning you can modify them after creating them. *Tuples* are immutable sequences, meaning they can't be changed after they're defined. *Strings* are a special type of sequence: each element in the sequence is a character.